Course Code: CS3331
Name of the Programme: B.Tech (Hons)
Name of the Course: Web Application Development
Semester: 6

Course credit	No. of hours per week (L + T + P)	Total no. of Teaching hours
3	2:0:2	30L + 30P

Course Objectives:

a) To deepen understanding of full stack web development using modern frontend and backend tools.

b) To build interactive, dynamic, and scalable web applications using React.js and state management.

b) To integrate NoSQL databases using MongoDB for data persistence.

c) To apply best practices for component-driven development and responsive design with Tailwind CSS.

e) To develop deployment-ready applications following modular architecture.

Module -1	No. of hours
	8

Modern Frontend Development with Tailwind CSS: -Introduction to utility-first CSS frameworks, Tailwind CSS vs Bootstrap, Setting up Tailwind CSS, Core concepts, Styling layout, typography, spacing, colors, buttons, and forms, Advanced utilities, Project structuring with reusable components, Hands-on: Build a responsive dashboard layout using Tailwind CSS.

Module - 2	No. of hours
	10

React.js Fundamentals and Component Architecture: -
Introduction to React, Functional components and props, State and lifecycle in function components (Hooks), Events, conditional rendering, and lists, Component communication, Project structure and reusable components, Hands-on: Build a React-based user interface with dynamic components

Module - 3	**No. of hours**
	12
State Management and Routing in React: - React Router, Global state management overview, Using Context API for state sharing across components, Redux fundamentals, Handling asynchronous actions with Redux Thunk, Debugging with React Developer Tools and Redux DevTools, Hands-on: Build a multi-page React application with context or Redux	

Module - 4	**No. of hours**
	15
Backend Integration and MongoDB with MERN Stack: - Overview of MERN stack architecture, Connecting frontend (React) to backend (Express) using Fetch/Axios, Introduction to MongoDB, Connecting to MongoDB using Mongoose ORM, Creating models and performing CRUD operations, API integration with React frontend and MongoDB backend, Hands-on: Build a complete data-driven MERN CRUD application.	

Module - 5	**No. of hours**
	15
Deployment, Authentication & Final Project: - Introduction to authentication with JWT and session tokens, Protected routes and role-based access in React and Express, Using environment variables for configuration, Deployment of	

frontend on Vercel/Netlify and backend on Render/Railway, Best practices for folder structure and production readiness, Final Project: Plan, build, test and deploy a complete full-stack web app.

Course outcomes
a) Design responsive UIs using Tailwind CSS and component-driven approach.
b) Develop interactive web apps using React.js and state management tools.
c) Build backend-integrated full stack apps with Express and MongoDB.
d) Deploy MERN stack applications and implement secure authentication.

Text Books	
1	"MERN Stack Front to Back" by Brad Traversy – Traversy Media
2	"The Road to React" by Robin Wieruch – Indepub
3	"You Don't Know JS (Book Series)" by Kyle Simpson – O'Reilly Media

Reference books	
1	Eric Elliott – Programming JavaScript Applications
2	Mosh Hamedani – Mastering React
3	Mark Erikson – Redux Essentials (Documentation)
4	MongoDB Docs – https://www.mongodb.com/docs/
5	React Official Docs – https://reactjs.org/docs/

Lab Programs /Tutorials	
1	Create a responsive layout using Tailwind CSS.
2	Build a dynamic component-based UI using React.
3	Implement routing and state management in a React application
4	Develop RESTful APIs using Express.js.
5	Perform CRUD operations using MongoDB and Mongoose ORM.
6	Integrate React frontend with Express backend using Axios.
7	Implement user authentication with JWT.
8	Deploy a full MERN stack project and present it with documentation.

Name of the Programme:	B.Tech (Hons)	
Title of the Course:	Web Application Development	
Course code	CS3331	
Semester:	VI	
Names of the Course Instructor(s)	Dr. V. Balaji	
Course credit	No. of hours per week (L + T + P)	Total no. of Teaching hours
3	2+0+2	30L+30P

Session	Module	Topic	Pedagogy / Activities	Reference
1	1	Introduction to Web Application Development	Interactive lecture	Course PPT
2	1	Full Stack Web Development Overview	Lecture + discussion	Course PPT
3	1	Utility-first CSS Frameworks	Interactive lecture	Tailwind Docs
4	1	Tailwind CSS vs Bootstrap	Comparative discussion	Tailwind Docs
5	1	Setting up Tailwind CSS	Live demonstration	Tailwind Docs
6	1	Tailwind Layout and Flex/Grid	Hands-on examples	Tailwind Docs
7	1	Typography, Spacing and Colors	Practice session	Tailwind Docs
8	1 (Lab)	Responsive Dashboard using Tailwind CSS	Coding + lab exercise	Lab Manual
9	2	Introduction to React and JSX	Interactive lecture	React Docs
10	2	Functional Components and Props	Board work + examples	React Docs
11	2	State and Hooks in React	Demonstration	React Docs
12	2 (Lab)	Building Dynamic React Components	Implementation	Lab Manual
13	2	Event Handling and Conditional Rendering	Problem-solving	React Docs
14	2	Lists and Component Communication	Guided coding	React Docs
15	2	Project Structure and Reusable Components	Discussion	Course PPT
16	2 (Lab)	React UI Mini Project	Coding + testing	Lab Manual
17	3	Introduction to React Router	Interactive lecture	React Router Docs
18	3	Routing and Navigation	Demonstration	React Router Docs
19	3	Global State Management Overview	Lecture	Course PPT
20	3 (Lab)	Context API Implementation	Hands-on coding	Lab Manual

21	3	Redux Fundamentals	Interactive lecture	Redux Docs
22	3	Asynchronous Actions with Redux Thunk	Walkthrough	Redux Docs
23	3	Debugging using DevTools	Demonstration	React DevTools
24	3 (Lab)	Multi-page React App with Redux	Implementation	Lab Manual
25	4	MERN Stack Architecture	Interactive lecture	Course PPT
26	4	Connecting React with Express	Demonstration	Express Docs
27	4	Introduction to MongoDB	Lecture	MongoDB Docs
28	4 (Lab)	MongoDB Connection using Mongoose	Coding	Lab Manual
29	4	CRUD Operations using MongoDB	Worked examples	MongoDB Docs
30	4	API Integration with React	Demonstration	Course PPT
31	4 (Lab)	MERN CRUD Application	Implementation	Lab Manual
32	4	Error Handling and Validation	Discussion	Course PPT
33	5	Authentication using JWT	Interactive lecture	JWT Docs
34	5	Protected Routes and Role-based Access	Demonstration	Course PPT
35	5	Environment Variables and Configuration	Lecture	Course PPT
36	5 (Lab)	Authentication Implementation	Coding	Lab Manual
37	5	Frontend Deployment (Vercel / Netlify)	Demo	Deployment Docs
38	5	Backend Deployment (Render / Railway)	Demo	Deployment Docs
39	5	Production Best Practices	Lecture	Course PPT
40	5 (Lab)	Full Stack Deployment	Hands-on	Lab Manual
41	5	Course Revision – Modules 1 & 2	Problem-solving	Course PPT
42	5	Course Revision – Modules 3 & 4	Discussion	Course PPT
43	5	Internal Assessment – I	Written test	Question Bank
44	5	Internal Assessment – II	Evaluation	Answer Scripts
45	5	Lab Internal Assessment	Practical exam	Lab Records
46	5	Security and Best Practices	Seminar	Reference Books
47	5	Emerging Trends in Web Development	Lecture	Recent Articles
48	5	Advanced MERN Use Cases	Tutorial session	Reference Books
49	5	Model Question Discussion	Guided discussion	Previous Papers
50	5	Mock Viva	Oral assessment	Viva Guidelines
51	5	Case Study Review	Student presentations	Student Work
52	5	Final Project Guidance	Faculty mentoring	Project Notes
53	5	Comprehensive Revision	Guided revision	Course PPT
54	5	Doubt Clearing Session	Open discussion	Course Notes
55	5	Pre-End Semester Lab Practice	Hands-on	Lab Manual
56	5	End Semester Theory Revision – I	Q&A session	Syllabus
57	5	End Semester Theory Revision – II	Problem-solving	Previous Papers
58	5	End Semester Lab Practice	Practical revision	Lab Manual
59	5	End Semester Viva Preparation	Mock viva	Viva Guidelines
60	5	Course Wrap-up and Reflection	CO discussion	Course PPT

Assessment and Evaluation

Continuous Internal Evaluation (CIE)	
Components of CIE	Weightage of each component
CIE 1 - Skilled Courses: Graded Component-1 - Quiz Questions Graded Component 2	10 marks out of 70 10 marks out of 70
CIE 2 Theory	25 marks out of 70
CIE 3 - Lab record writeup - Lab Project & Viva	5 marks out of 70 20 marks out of 70
Total	70

Semester End Exams (SEE)	
Mode of Exam	Theory
Weightage of exam	30

Course Outcomes- Progamme Outcomes Matrix

Sl. No.	Course Outcome (CO)	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8
1	Design responsive and user-friendly web interfaces using Tailwind CSS	3	2	2	–	–	–	–	–
2	Develop interactive frontend applications using React.js and component-based architecture	3	3	3	1	–	–	–	–
3	Implement state management and routing in single-page applications	2	3	3	2	–	–	–	–
4	Build backend-integrated full-stack applications using Express and MongoDB	2	3	2	3	2	–	–	–
5	Deploy secure MERN stack applications with authentication and best practices	2	3	3	3	3	–	–	–

Rubrics for evaluation of CIE (separate for each CIE component)

<table>
<tr><td>Programme Outcome</td><td>PO1 – Computing Fundamentals
Apply knowledge of computing fundamentals, programming concepts, data structures, and web technologies for application development.

PO2 – Problem Analysis
Identify, analyze, and solve well-defined web application development problems using appropriate frontend and backend technologies.

PO3 – Design and Development of Solutions
Design and develop scalable, responsive, and interactive web applications using modern frameworks, component-based architecture, and full-stack approaches.

PO4 – Modern Tool Usage
Use modern web development tools, frameworks, libraries, and deployment platforms such as React, Tailwind CSS, Express, MongoDB, and cloud services effectively.

PO5 – Communication Skills
Communicate design ideas, implementation details, and project outcomes effectively through documentation, presentations, and demonstrations.

PO6 – Teamwork and Leadership
Function effectively as an individual or as a member of a team in web application development projects and laboratory environments.

PO7 – Professional Ethics
Apply ethical principles, security practices, accessibility standards, and social responsibility in the development and deployment of web applications.

PO8 – Lifelong Learning
Engage in self-directed and continuous learning to adapt to emerging web technologies, frameworks, and industry practices.</td></tr>
<tr><td>Total Marks</td><td>70</td></tr>
<tr><td colspan="2"></td></tr>
</table>

CIE components		CO-PO	Excellent (5)	Very Good (4)	Good (3)	Average (2)	Needs Improvement (1)
CIE -1 Skills Build Courses: Graded Component- Weightage: 10 **Graded Component 1: Quiz** *(Conceptual and theoretical understanding of Web Application Development fundamentals – HTML, CSS, Tailwind, React, MERN architecture)*		CO1, CO2 PO1, PO2	Answers all questions accurately with clear understanding; demonstrates excellent conceptual clarity	Minor errors; understands most concepts clearly	Moderate understanding; some confusion in key concepts	Many errors; shows poor understanding	understanding No attempt or completely incorrect
CIE -1 Weightage: 10 **Graded Component 2: Debugging / Code Completion Task** *(Hands-on correction and completion of frontend/backend code using React components, Tailwind CSS layouts, Express APIs, or MongoDB CRUD operations)*		CO1, CO2 PO1, PO3	Identifies and corrects all errors; completes code with proper structure, reusable components, and best practices	Minor issues; mostly correct logic and structure	Code works with basic functionality but lacks clean structure or optimization	Many syntax or logical issues; partially functional code	No submission or non-functional code
CIE-2 Mid Sem Examination (Theory) Weightage: 25		CO1, CO2 CO3, CO4, CO5 PO1, PO2, PO3					

Sl.no	Content
Part A – 5 Marks *(Minimum 1 mark, Max 2 marks, Max 5 Questions)*	5 Questions of 1 mark
Part B – 20 Marks *(Compulsory 4 questions of 5 marks each, there can be max 3 sub-divisions)*	4 Questions of 5 marks with 2 or 3 sub-divisio possibly

		, PO4 , PO5	
CIE-3 Weightage: 25 **Lab Exercise + Record** Weightage: 10		CO1 , CO2 CO3 , CO4 , CO5 PO1 , PO2 , PO3 , PO4 , PO5	(rubric below)

Sl. No	Criteria	Measuring Method	Excellent (Full Marks)	Good (Partial Marks)	Poor (Zero Marks)	Marks
1	**Understanding of Problem Statement**	Observations	Demonstrates clear understanding of the web application problem and correctly identifies appropriate frontend and backend technologies, architecture, and design approach.	Shows partial understanding; selects some appropriate technologies or design approaches but with minor gaps.	Lacks clarity or fails to identify suitable web technologies or application design.	
2	**Implementation / Execution of Code**	Observations	Successfully implements the web application with correct logic, component structure, API integration, and expected output; code executes without errors and follows	Implements the application partially; output is obtained but with logical errors, poor structure, or incomplete integration.	Major logical errors or non-functional application.	

						best practices.				
			3	**Results and Documentation**	Observations	Produces correct results with functional UI, proper API responses, well-commented code, screenshots (if applicable), and clearly organized documentation.	Results are produced but documentation, screenshots, or code comments are incomplete or unclear.	No output produced or documentation is missing.		
VIVA- 5		CO1, CO4 PO1, PO2	**Sl. No**	**Criteria**	**Method**	**Excellent (4 M)**	**Good (3 M)**	**Poor (0 M)**	**Marks**	**Remarks**
			1	**Conceptual Understanding**	Viva Voce	Explains thoroughly the core web application development concepts, frameworks, and principles (such as Tailwind CSS, React	Adequately explains the web development concepts used in the lab with reasonable clarity and partial correctness.	Unable to explain the core web development concepts used in the lab activity or gives incorrect / irrelevant answers.	5	

				components, state management, backend integration, or deployment) applied in the corresponding lab task.		
CIE-3 Lab Project Weightage: 10	CO1, CO2, CO3, CO4 PO1 - PO8	Code is fully correct, optimized, modular, and well-commented; demonstrates clear understanding of frontend and backend web technologies, proper component structure, API integration, and effectively handles edge cases.	Minor mistakes or inefficiencies; logic is mostly correct; component structure and integration are largely sound, with a few edge cases missed.	Basic web application logic is implemented; functionality works but contains errors, poor structure, or incomplete handling of states, inputs, or responses.	Major logical or structural issues; application partially runs but is incomplete, poorly organized, or error-prone.	No submission, non-functional application, or irrelevant work.

Signature of the Course Faculty